English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

# Probability and Statistics, Computer Science

Regular exam, 2 July 2023: solutions

*English transcription of the supplied handwritten workings. The source's numerical conclusions and noted ambiguities are retained.*

## Problem 1

### 1. Percentage measurement error

$$\mathbb{P}(\text{error} > 10\%) = \mathbb{P}(N(0,5^2) > 10) = \mathbb{P}(N(0,1) > 2) = 1 - \Phi(2) =: p \approx 2.28\%.$$

The expected number of cars until the first such error is $\mathbb{E}[Ge(p)] = 1/p$. Among 1000 cars, the expected number with error above 10% is $\mathbb{E}[Bin(1000, p)] = 1000p$. For iid $X_i \sim N(0, 5^2)$,

$$\begin{aligned}\mathbb{P}\left(\frac{X_1 + \cdots + X_{100}}{100} > 1\right) &= \mathbb{P}(N(0, 100 \cdot 5^2) > 100) \\ &= \mathbb{P}(N(0,1) > 2) = 1 - \Phi(2) \approx 2.28\%.\end{aligned}$$

### 2. Car speed and recorded speed

Let $V \sim N(40, 5^2)$ be speed and $X \sim N(0, 5^2)$ instrument error. Then $\mathbb{P}(V > 50) = \mathbb{P}(N(0,1) > 2) \approx 2.28\%$. The notes point out that the actual question asks for the probability that the instrument reports more than 50 km/h:

$$\mathbb{P}\big(V(1 + X/100) > 50\big) = \mathbb{P}\left(N_1(40, 5^2)\, N_2\left(1, (1/20)^2\right) > 50\right).$$

It is difficult to evaluate explicitly; either $\mathbb{P}(V > 50)$ or the expression above was accepted. A Monte Carlo estimate using $10^6$ simulations is about 3.46%.

For the mean absolute speed error represented by signed products, the notes seek $\sigma$ from

$$\mathbb{P}\left(\frac{X_1V_1 + \cdots + X_{100}V_{100}}{10000} > 1\right) \le 0.1\%.$$

Independence gives

$$\mathbb{E}(X_1V_1) = \mathbb{E}X_1\mathbb{E}V_1 = 0, \qquad D(X_1V_1) = \mathbb{E}X_1^2\mathbb{E}V_1^2 - (\mathbb{E}X_1)^2(\mathbb{E}V_1)^2 = \sigma^2(40^2 + 25).$$

The CLT approximation is

$$\mathbb{P}\left(N(0,1) > \frac{10000}{\sigma\sqrt{1625}\sqrt{100}}\right).$$

The handwritten calculation sets the threshold approximately equal to 3.72 and obtains $\sigma \approx 6.67$. It comments that the exam wording was ambiguous and accepts any correct calculation with a comparable interpretation.

### 3. Absolute errors

For $|N(0, 5^2)|$, the notes use $p = 2\Phi(-2)$ and repeat the geometric and binomial arguments. For the average absolute error they write

$$\mathbb{P}\left(\frac{|X_1| + \cdots + |X_{100}|}{100} > 1\right) \approx \mathbb{P}\left(N(0,1) > \frac{100 - 100\mathbb{E}|X_1|}{\sqrt{100D|X_1|}}\right).$$

The displayed moment calculation is

$$\mathbb{E}|X_1| = 5\mathbb{E}|N(0,1)| = 5 \cdot 2\int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-x^2/2}\, dx = \frac{5}{\sqrt{2\pi}},$$

followed by $D|X_1| = 25 - 25/(2\pi)$ and "substitute". *Transcription note: the last equality for the absolute normal mean drops a factor of 2 in the handwritten source. Its following variance and numerical threshold above are preserved, rather than silently replaced.*

# Problem 2

## 1. Independence of the counts

Let $I = i_1 + \cdots + i_n$ and $F_n = f(1) + \cdots + f(n)$. Conditioning on $N$ gives

$$\begin{aligned}
&\mathbb{P}(X_1 = i_1, \ldots, X_n = i_n) \\
&= \sum_{k=I}^{\infty} \mathbb{P}(N = k)\mathbb{P}(X_1 = i_1, \ldots, X_n = i_n \mid N = k) \\
&= \sum_{k=I}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} \frac{k!}{i_1! \cdots i_n!(k-I)!} \prod_{j=1}^{n} f(j)^{i_j} (1 - F_n)^{k-I} \\
&= e^{-\lambda} \prod_{j=1}^{n} \frac{(\lambda f(j))^{i_j}}{i_j!} \sum_{r=0}^{\infty} \frac{\lambda^r (1 - F_n)^r}{r!} \\
&= \prod_{j=1}^{n} \frac{e^{-\lambda f(j)} (\lambda f(j))^{i_j}}{i_j!}.
\end{aligned}$$

Thus the $X_i$ are independent and $X_i \sim Poi(\lambda f(i))$.

## 2. Expected sales until stopping

The event $\{T \le n-1\}$ depends only on $X_1, \ldots, X_{n-1}$, so it is independent of $X_n, X_{n+1}, \ldots$. Its complement $\{T \ge n\}$ is therefore independent of $X_n$. Consequently,

$$\begin{aligned}
\mathbb{E}S = \mathbb{E}\sum_{i=1}^{\infty} X_i \mathbb{1}_{\{T \ge i\}} &= \sum_{i=1}^{\infty} \mathbb{E}X_i \mathbb{P}(T \ge i) \\
&= \sum_{i=1}^{\infty} \mathbb{E}X_i \sum_{j=i}^{\infty} \mathbb{P}(T = j) = \sum_{j=1}^{\infty} \mathbb{P}(T = j) \sum_{i=1}^{j} \mathbb{E}X_i \\
&= \lambda \sum_{j=1}^{\infty} \mathbb{P}(T = j)(f(1) + \cdots + f(j)) = \lambda \mathbb{E}[f(1) + \cdots + f(T)].
\end{aligned}$$

*Warning in the source:* starting instead from

$$\mathbb{E}S = \sum_{k \ge 1} \mathbb{P}(T = k)\mathbb{E}[X_1 + \cdots + X_k \mid T = k],$$

one cannot replace the conditional expectation by $\mathbb{E}X_1 + \cdots + \mathbb{E}X_k$, since $\{T = k\}$ is an event determined by $X_1, \ldots, X_k$ and is not independent of them.

# Problem 3

## 1. Fractional parts

Let $X = \{MU\}$. For $t \in (0, 1)$,

$$\mathbb{P}(X \le t) = \sum_{k=0}^{M-1} \mathbb{P}(MU \in [k, k + t]) = M \frac{t}{M} = t,$$

since $MU \sim U(0, M)$. Hence $X \sim U(0, 1)$, $\mathbb{E}X = 1/2$ and $DX = 1/12$. To compute correlation,

$$Cor(X, U) = \frac{\mathbb{E}[XU] - 1/4}{1/12}.$$

Partitioning the unit interval,

$$\begin{aligned}
\mathbb{E}[XU] &= \sum_{k=0}^{M-1} \int_{k/M}^{(k+1)/M} (Mt-k)t\,dt \\
&= M\int_0^1 t^2 dt - \sum_{k=1}^{M-1} k \int_{k/M}^{(k+1)/M} t\,dt \\
&= \frac{M}{3} - \frac{1}{2M^2}\sum_{k=1}^{M-1}(2k^2+k) \\
&= \frac{M}{3} - \frac{1}{2M^2}\left[2\frac{(M-1)M(2M-1)}{6} + \frac{(M-1)M}{2}\right] \\
&= \frac{1}{12M}\left[4M^2-(M-1)(4M+1)\right] = \frac{1}{12}\left(3+\frac{1}{M}\right).
\end{aligned}$$

Thus $Cor(X,U) = 1/M$. The larger $M$, the less correlated $\{MU\}$ and $U$ are.

## 2. Extremes of 100 uniforms

Each $U_i$ has mean $1/2$. For $0 < t < 1$,

$$\mathbb{P}(\max(U_1,\ldots,U_{100}) \le t) = \mathbb{P}(U_1 \le t)^{100} = t^{100}.$$

The maximum has density $100t^{99}$ and expectation $\int_0^1 100t^{100}dt = 100/101$. Since $1-U_i \overset{d}{=} U_i$,

$$\min(U_1,\ldots,U_{100}) \overset{d}{=} 1 - \max(U_1,\ldots,U_{100}),$$

so its expectation is $1/101$. Alternatively, derive its distribution function directly.

## 3. Tangent of a uniform angle

For $V \sim U(-\pi/2, \pi/2)$ and $Y = \tan V$,

$$\mathbb{P}(Y \le t) = \mathbb{P}(V \le \arctan t) = \int_{-\pi/2}^{\arctan t} \frac{1}{\pi}dx = \frac{\arctan t + \pi/2}{\pi}.$$

Hence $f_Y(t) = 1/(\pi(1+t^2))$ for $t \in \mathbb{R}$. The integrand $t/(\pi(1+t^2))$ behaves as $1/(\pi t)$ at infinity and is not integrable, so $\mathbb{E}Y$ and $DY$ are not defined.

## 4. Cauchy transformations

Write $Z \overset{d}{=} \tan V$ as above. Then

$$1/Z \overset{d}{=} 1/\tan V = \cot V \overset{d}{=} \tan V,$$

with the instruction "formalise why!". Also

$$\frac{2\tan V}{1-\tan^2 V} = \tan(2V) \overset{d}{=} \tan V, \qquad \frac{3\tan V - \tan^3 V}{1-3\tan^2 V} = \tan(3V) \overset{d}{=} \tan V.$$

The usual density-transformation method is another option.

# Bonus

1. See the law of total variance.
2. See Wikipedia; by symmetry $\mathbb{E}X = 1/2$, and $DX = 1/8$.
3. The derivative is zero almost everywhere, so $X$ has no density. The Cantor function is an example of a continuous increasing function that is not absolutely continuous.